



EntryLevel

DATA ANALYST: SQL PORTFOLIO

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Professional Background

I'm a detail-oriented aspiring Data Analyst looking to further my knowledge and experience.

I'm in pursuit of a new career in data analytics. Holding a Bachelors Degree in Music Business, I have recently pivoted into tech for better opportunities.

In my last position, I worked with Salesforce and I'm working on diving deeper into the analytical side in the future.

I've recently stepped into a new position working with data such as collecting, importing, and cleaning. I bring a set of valuable talents such as my analytical mind, organization, and strategic thinking. I have a knack for problem-solving and research.

I've been working on upskilling not only with data analytics, but also with project management to help my organization and leadership skills. I recently obtained a ScrumMaster Certification, a Jira Fundamentals badge, as well as gained more knowledge in Agile framework. I'm highly dedicated to growing as a data professional and will continue to learn more in hopes of landing the right job opportunity.

Thank you for your time and consideration.

Thank you

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Introduction

As a Data Analyst working for the charity, **Education for All**, I was asked by the Head of Fundraising to present data on donor insights and donation rates in order to increase donations next year during my team's fundraising strategy meeting in two weeks.

The objectives of this report are to:

- Increase the number of donors in your database
- Increase the donation frequency of your donors.
- Increase the value of donations in your database.

I first had to analyze the root cause. I then collected data such as donor name, gender, donation amounts, donation frequency, state, job field, university, etc that were provided to me in two dataset:

EFO_Donation_Data and EFO_Donor_Data. I ran both datasets in SQL and Tableau to analyze and visualize the data and find answers to the business problem.

This report will show which gender donates most often, highest amounts donated, which states our highest and lowest donations come from, and more.

Root Cause Analysis

Root Cause Analysis is the process used to find what the root cause/underlying problem is in order to figure out how to solve the problem.

- Why don't we have enough donors?
- Why are our donors not donating frequently?
- Why are our donors not making large donations?
- Why don't we have more pertinent information about our donors in our database?
- Why are males donating higher amounts than females?

The root cause of the problem is that we don't have enough donations. We want to increase donation amounts, as well as frequency in order for the organization to continue to grow.



Insights SQLite

I was given 2 datasets that I imported into SQLite and Tableau, as shown below.

EFO_Donation_Data	EFO_Donor_Data
Contains the following data	Contains the following data
id Donor ID	id Donor ID
first_name Donor first name	donation_frequency
last_name Donor last name	Frequency of donation
email Donor email address	university Donor University attended
gender Donor gender	car Donor car make
job_field Donor job field	second_language Donor second language
donation Donation amount	favourite_colour Donor favourite colour
state Donor state of residence (US)	movie_genre Donor favourite movie genre
shirt_size = Donor t-shirt size	

SUM() was used to retrieve the total sum of donations:

```
1 SELECT sum(donation)
2 FROM Donation_Data;
```

COUNT() was used to find the number of donors grouped by gender:

```
1 SELECT gender, COUNT(*)
2 FROM Donation_Data
3 GROUP BY gender
4 ORDER BY COUNT(*) DESC;
```

To find the sum of donations grouped by gender, the following queries were used:

```
1 SELECT gender, sum(donation)
2 FROM Donation_Data
3 GROUP BY gender
4 ORDER BY sum(donation) DESC;
```

Insights

Top 5 states with highest sum of donations:

```
1 SELECT state, sum(donation)
2 FROM Donation_Data
3 GROUP BY state
4 ORDER BY sum(donation) DESC
5 LIMIT 5;
```

Top 5 states with lowest sum of donations:

```
1 SELECT state, sum(donation)
2 FROM Donation_Data
3 GROUP BY state
4 ORDER BY sum(donation) ASC
5 LIMIT 5;
```

The LEFT JOIN query was used to find the donation frequency by gender, grouped by job field.

Females:

```
1 SELECT Donation_Data.gender, Donor_Data2.donation_frequency, Donation_Data.job_field, COUNT(*)
2 FROM Donation_Data
3 LEFT JOIN Donor_Data2
4 ON Donation_Data.id = Donor_Data2.id
5 WHERE donation_frequency = 'Once'
6 AND gender = 'Female'
7 GROUP BY job_field
8 ORDER BY COUNT(*) DESC;
```

```
1 SELECT Donation_Data.gender, Donor_Data2.donation_frequency, Donation_Data.job_field, COUNT(*)
2 FROM Donation_Data
3 LEFT JOIN Donor_Data2
4 ON Donation_Data.id = Donor_Data2.id
5 WHERE donation_frequency = 'Weekly'
6 AND gender = 'Female'
7 GROUP BY job_field
8 ORDER BY COUNT(*) DESC;
```

```
1 SELECT Donation_Data.gender, Donor_Data2.donation_frequency, Donation_Data.job_field, COUNT(*)
2 FROM Donation_Data
3 LEFT JOIN Donor_Data2
4 ON Donation_Data.id = Donor_Data2.id
5 WHERE donation_frequency = 'Monthly'
6 AND gender = 'Female'
7 GROUP BY job_field
8 ORDER BY COUNT(*) DESC;
```

```
1 SELECT Donation_Data.gender, Donor_Data2.donation_frequency, Donation_Data.job_field, COUNT(*)
2 FROM Donation_Data
3 LEFT JOIN Donor_Data2
4 ON Donation_Data.id = Donor_Data2.id
5 WHERE donation_frequency = 'Yearly'
6 AND gender = 'Female'
7 GROUP BY job_field
8 ORDER BY COUNT(*) DESC;
```

Insights

Males:

```
1 SELECT Donation_Data.gender, Donor_Data2.donation_frequency, Donation_Data.job_field, COUNT(*)
2 FROM Donation_Data
3 LEFT JOIN Donor_Data2
4 ON Donation_Data.id = Donor_Data2.id
5 WHERE donation_frequency = 'Once'
6 AND gender = 'Male'
7 GROUP BY job_field
8 ORDER BY COUNT(*) DESC;
```

```
1 SELECT Donation_Data.gender, Donor_Data2.donation_frequency, Donation_Data.job_field, COUNT(*)
2 FROM Donation_Data
3 LEFT JOIN Donor_Data2
4 ON Donation_Data.id = Donor_Data2.id
5 WHERE donation_frequency = 'Weekly'
6 AND gender = 'Male'
7 GROUP BY job_field
8 ORDER BY COUNT(*) DESC;
```

```
1 SELECT Donation_Data.gender, Donor_Data2.donation_frequency, Donation_Data.job_field, COUNT(*)
2 FROM Donation_Data
3 LEFT JOIN Donor_Data2
4 ON Donation_Data.id = Donor_Data2.id
5 WHERE donation_frequency = 'Monthly'
6 AND gender = 'Male'
7 GROUP BY job_field
8 ORDER BY COUNT(*) DESC;
```

```
1 SELECT Donation_Data.gender, Donor_Data2.donation_frequency, Donation_Data.job_field, COUNT(*)
2 FROM Donation_Data
3 LEFT JOIN Donor_Data2
4 ON Donation_Data.id = Donor_Data2.id
5 WHERE donation_frequency = 'Yearly'
6 AND gender = 'Male'
7 GROUP BY job_field
8 ORDER BY COUNT(*) DESC;
```

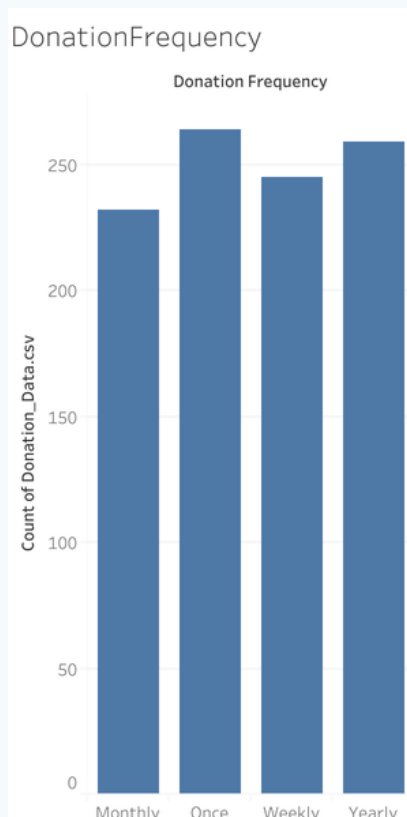
I also wanted to take a look at how many of our donors do not have university education by using the following query where donors showed NULL under the university column.

```
1 SELECT university, COUNT(donation)
2 FROM Donation_Data
3 LEFT JOIN Donor_Data2
4 ON Donation_Data.id = Donor_Data2.id
5 WHERE university IS NULL
6 GROUP BY university
7 ORDER BY COUNT(donation) DESC;
```

...and how many of those without university education have donated \$400 and more:

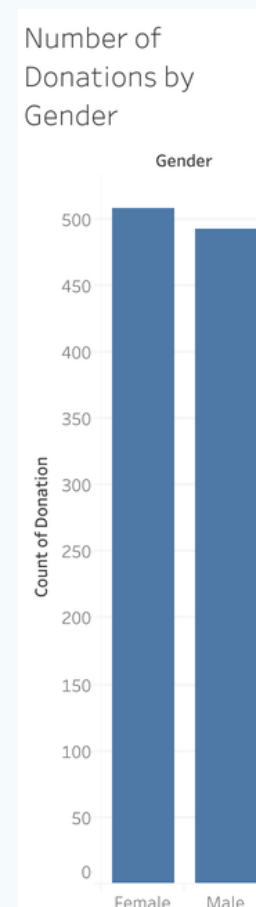
```
1 SELECT Donation_Data.gender, Donation_Data.donation, Donor_Data2.university
2 FROM Donation_Data
3 LEFT JOIN Donor_Data2
4 ON Donation_Data.id = Donor_Data2.id
5 WHERE university IS NULL
6 AND donation >= 400
7 ORDER BY donation DESC;
```

Insights Tableau

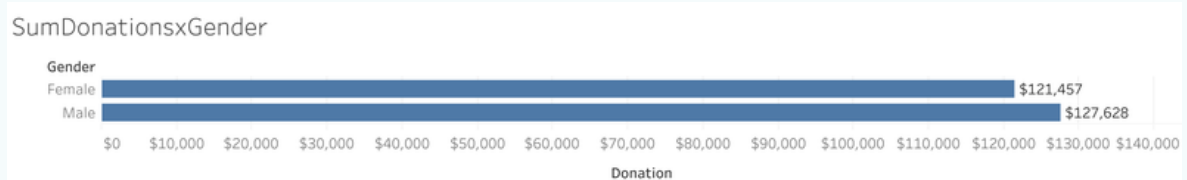


Left chart: Majority of our donors have only donated once and are not active donors.

Right chart: Our female donors have made more donations.

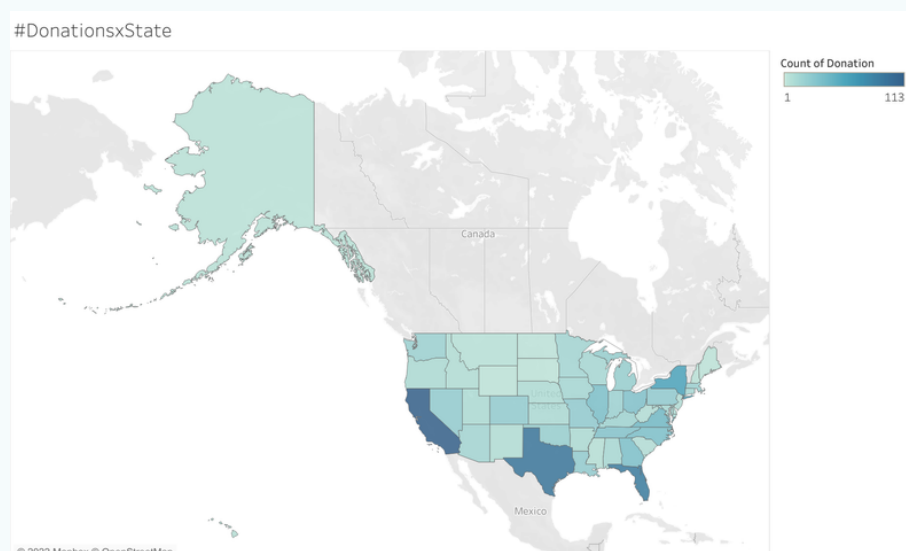


However, our male donors have donated a higher sum of donations.



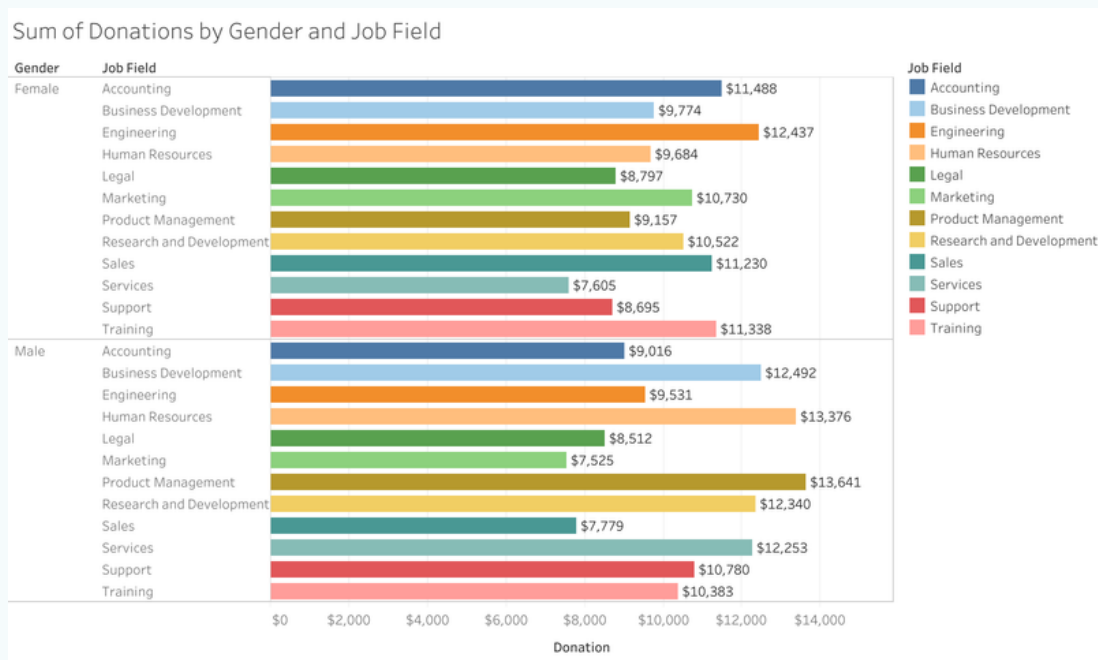
Top 5 donor states are California, Texas, Florida, New York, and Virginia.

The lowest are Maine, South Dakota, Wyoming, North Dakota, and Alaska.,

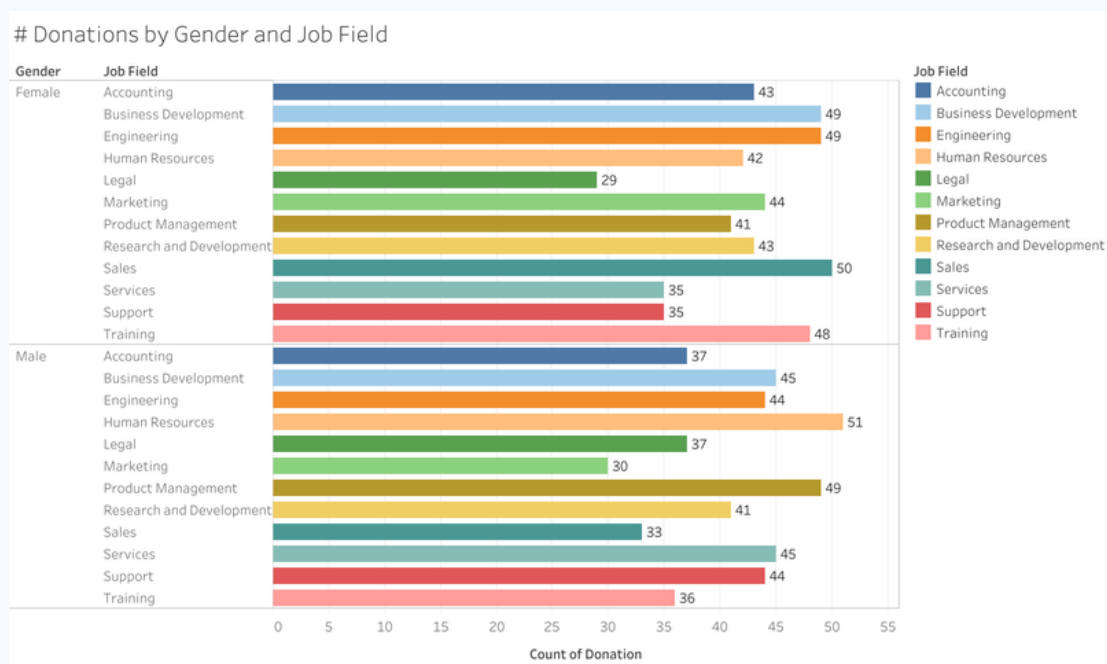


Insights

The sum of donations by gender and job field show that our female donors in Engineering have donated the highest value. Our male donors in Product Management have donated the highest value.



However, more donations have come from our female donors in Sales and male donors in Human Resources.



Findings and Recommendations

My analysis shows that:

- The total sum of donations is \$249,085
- Male donors have donated \$127,628 and female donors \$121,457 of that total sum. However, we have more female donors than male.
- 748 of our donors have university education. They have donated \$187,456.
- 252 of our donors do not have university education and have donated \$61,629.
- The majority of our donors have only donated once and yearly. It would benefit us to reach out to our one-time donors and see if they'd like to donate again

donation	donation_frequency	sum(donation)
292	Once	\$64,586
255	Yearly	\$65,667
178	Monthly	\$59,680
28	Weekly	\$59,152

Top 5 states with highest donation sum.

state	sum(donation)
California	\$30,264
Texas	\$24,097
Florida	\$20,562
New York	\$14,759
Virginia	\$10,750

We should reach out to all of these states to see if they'd like to donate more. But try to attract other states.

Top 5 states with lowest donation sum.

state	sum(donation)
Wyoming	\$232
Maine	\$258
South Dakota	\$401
North Dakota	\$651
Alaska	\$734

Findings and Recommendations

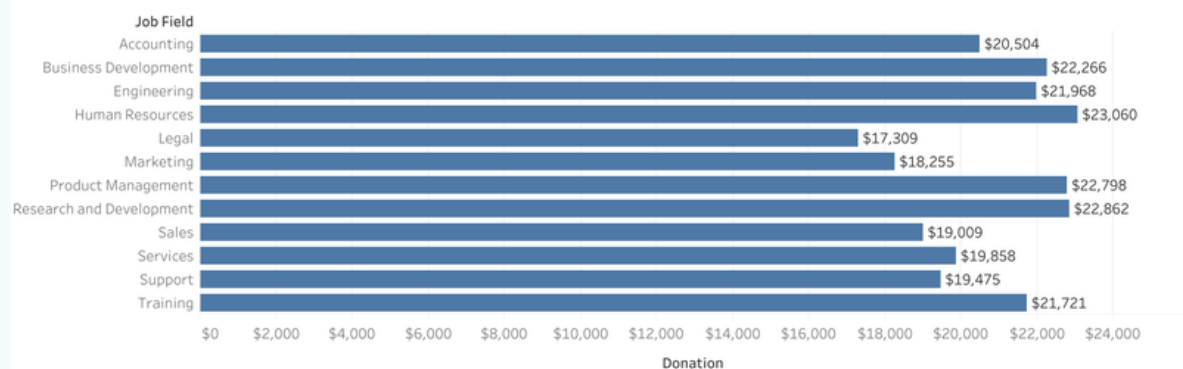
I recommend reaching out to our top 10 female and male inactive donors (one-time donors) to see why they haven't donated again and if they'd like to.

first_name	donation	donation_frequency	gender	state	job_field	university
Geoffry	422	Once	Female	California	Marketing	null
Odelle	420	Once	Female	New York	Marketing	Grasha
Laure	364	Once	Female	Kansas	Human Resources	Abthorp
Emelda	201	Once	Female	Illinois	Human Resources	Denkin
Isabella	299	Once	Female	Kentucky	Legal	Panks
Gerhardine	287	Once	Female	Oregon	Product Management	null
Tobie	286	Once	Female	Indiana	Accounting	null
Kellina	482	Once	Female	Massachusetts	Support	Mellonby
Laney	481	Once	Female	Arkansas	Services	Khrstoforov
Aprillette	468	Once	Female	Michigan	Accounting	null

first_name	donation	donation_frequency	gender	state	job_field	university
Hanson	482	Once	Male	California	Training	Ellerey
Saw	468	Once	Male	Oregon	Legal	Gorler
Sven	451	Once	Male	Iowa	Accounting	Blount
Anastasi	400	Once	Male	New York	Sales	null
Mendel	196	Once	Male	North Carolina	Human Resources	Mitchiner
Chrissie	341	Once	Male	Illinois	Research and Development	Ennion
Emmit	491	Once	Male	Nevada	Training	McIlmorow
Padraig	488	Once	Male	Illinois	Product Management	null
Shanie	486	Once	Male	Florida	Business Development	Brooke
Barbara-anne	470	Once	Male	District of Columbia	Product Management	Thams

I also recommend reaching out to our donors in Human Resources, Research and Development, Product Management, and Business Development. Our donors from these job fields have donated the largest total sum.

SumDonationsxJobField



Conclusion

To conclude my analysis, after importing and joining both datasets, EFO_Donation_Data and EFO_Donor_Data, I have found ways to:

- Increase the number of donors in your database
- Increase the donation frequency of your donors.
- Increase the value of donations in your database.

It appears that although we have more female donors than males, our male donors have donated the highest total of donation value. This may be because our male donors work in higher paying job fields or states, or even higher education background. However, we should reach out and attract more females working in Engineering and Sales and males in Product Management and Human Resources, as they have shown to donate more value and most of our donors are in those groups.

My analysis has also shown me that most of our donors donate yearly and once. We especially want to reach out to our one-time donors to see if they'd like donate again or more frequently in order to improve our donation frequency.

The majority of our donors who also donate the highest values are coming from states with bigger cities and population such as California, Texas, Florida, New York, and Virginia. We should be reaching out to the states mentioned as well as to the smaller states to see if they've heard of our organization and would like to donate in order to increase our donation value and number of donors.

I've also analyzed that approximately 75% of our donors have university education. I recommend promoting our organization to more people with university education.

Lastly, I recommend collecting more pertinent information about our donors, as data like car, shirt size, second language, favorite color and movie genre did not matter in finding answers to our business problem.